Agent Generated Hypothesis and Research Plan

Novel Hypothesis:

Metformin, in combination with a ketogenic diet, will synergistically inhibit GBM growth, particularly in subtypes with high OXPHOS dependence, by disrupting mitochondrial metabolism and inducing oxidative stress, leading to increased apoptosis in GBM cells. This combination may enhance the efficacy of radiation therapy, as suggested by existing clinical trials exploring similar combinations.

Research Plan:

This plan outlines in vitro and in vivo experiments to validate the hypothesis.

- **In vitro studies:**
- **Cell lines:** Select a panel of GBM cell lines representing different molecular subtypes (e.g., proneural, mesenchymal, classical). Characterize their metabolic profiles, including OXPHOS dependence, using Seahorse XF analyzer. This is crucial for identifying cell lines that exhibit varying degrees of dependence on oxidative phosphorylation (OXPHOS).
- **Treatment:** Treat the cell lines with metformin, ketogenic diet (using beta-hydroxybutyrate or a similar ketone body), radiation, and combinations thereof. The dosages should be optimized based on literature and preliminary dose-response experiments.
- **Assays:**
- Cell viability assays (e.g., MTT, CellTiter-Glo) to assess the cytotoxic effects of the treatments after 24, 48, and 72 hours.
- Apoptosis assays (e.g., flow cytometry with Annexin V/PI staining) to measure the induction of apoptosis at different time points (e.g., 24, 48 hours).
- Mitochondrial respiration assays (Seahorse XF analyzer) to evaluate the impact on mitochondrial function, measuring oxygen consumption rate (OCR) and extracellular acidification rate (ECAR).
- Measurement of oxidative stress markers (e.g., ROS levels using DCFDA assay) to quantify reactive oxygen species production.
- Western blotting to assess changes in key signaling pathways (e.g., AMPK, mTOR, Akt, cleaved caspase-3).
- **In vivo studies:**
- **Animal model:** Use orthotopic GBM xenograft models in immunocompromised mice (e.g., NSG mice), using GBM cell lines that showed promising results in vitro, especially those with high OXPHOS dependence. Inject a defined number of cells (e.g., 1x10^5 cells) into the brain.
- **Treatment:** Administer metformin (e.g., via intraperitoneal injection), ketogenic diet (formulated rodent chow), radiation, and combinations thereof to the mice. Start treatment 7-10 days post-implantation when tumors are established.
- **Monitoring:**
- Monitor tumor growth using MRI weekly.
- Assess survival rates by Kaplan-Meier survival analysis.

- Perform immunohistochemistry on tumor tissues to evaluate apoptosis (e.g., cleaved caspase-3 staining), proliferation (e.g., Ki-67 staining), and changes in signaling pathways (e.g., p-AMPK, p-mTOR).
- **Subtype stratification:** Analyze publicly available GBM datasets (e.g., TCGA) to identify potential biomarkers (gene expression signatures, mutations) that predict sensitivity to metformin and ketogenic diet based on metabolic profiles. This can involve correlation analysis between gene expression related to OXPHOS and clinical outcomes in patients treated with similar therapies (if available).

Novelty Assessment:

Based on the PubMed searches performed (specifically, the search for "metformin glioblastoma ketogenic diet metabolism"), the core hypothesis of combining metformin and a ketogenic diet for GBM treatment is not entirely novel. One article (PMID: 34152528) describes a Phase I clinical trial exploring a similar combination with radiation therapy. However, the specific focus on identifying and exploiting metabolic vulnerabilities within GBM subtypes, particularly high OXPHOS dependence, to enhance the efficacy of this combination remains a relatively unexplored area. The hypothesis that subtypes with higher OXPHOS dependence will be more sensitive to this combination adds a layer of refinement and potential novelty. The proposed research plan, particularly the inclusion of metabolic characterization of cell lines and analysis of publicly available datasets to identify predictive biomarkers, aims to further differentiate this research from existing studies.